

Q1 - 24 June - Shift 1

Let $\hat{a}, \hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and

$\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$, then $|6\vec{c}|^2$ is equal to

(A) $6(3 - \sqrt{3})$

(B) $3 + \sqrt{3}$

(C) $6(3 + \sqrt{3})$

(D) $6(\sqrt{3} + 1)$

Space for your notes:

Q2 - 24 June - Shift 1

If the shortest distance between the line

$$\vec{r} = (-\hat{i} + 3\hat{k}) + \lambda(\hat{i} - a\hat{j}) \text{ and}$$

$$\vec{r} = (-\hat{j} + 2\hat{k}) + \mu(\hat{i} - \hat{j} + \hat{k}) \text{ is } \sqrt{\frac{2}{3}}, \text{ then the integral}$$

value of a is equal to

Space for your notes:

Q3 - 24 June - Shift 2

Questions

MathonGo

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that

$$\left| (\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b}) \right| = 2. \text{ If } \theta \in (0, \pi) \text{ is the angle}$$

between $\hat{a}$ and $\hat{b}$, then among the statements :

$$(S1) : 2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$$

$$(S2) : \text{The projection of } \hat{a} \text{ on } (\hat{a} + \hat{b}) \text{ is } \frac{1}{2}$$

(A) Only (S1) is true

(B) Only (S2) is true

(C) Both (S1) and (S2) are true

(D) Both (S1) and (S2) are false

Q4 - 25 June - Shift 1

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0, i = 1, 2, 3$ be a vector which makes equal angles with the coordinates axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let

$\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90° .

If $\vec{a}, \vec{b}$ and x-axis are coplanar, then projection

of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to

$$(A) \sqrt{7}$$

$$(B) \sqrt{2}$$

$$(C) 2$$

$$(D) 7$$

Q5 - 25 June - Shift 1

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Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$,

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where $|\vec{a}| = 4$, $|\vec{b}| = 3$ $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then

$\left|(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})\right|^2 + 4(\vec{a} \cdot \vec{b})^2$ is equal to _____

Q6 - 25 June - Shift 2

Let $\vec{b} = \hat{i} + \hat{j} + \lambda \hat{k}$, $\lambda \in \mathbb{R}$. If $\vec{a}$ is a vector such that

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$\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then

$(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ is equal to

Q7 - 26 June - Shift 1

If $\vec{a} \cdot \vec{b} = 1$, $\vec{b} \cdot \vec{c} = 2$ and $\vec{c} \cdot \vec{a} = 3$, then the value

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of $\left[\vec{a} \times (\vec{b} \times \vec{c}), \vec{b} \times (\vec{c} \times \vec{a}), \vec{c} \times (\vec{b} \times \vec{a})\right]$ is :

- (A) 0 (B) $-6\vec{a} \cdot (\vec{b} \times \vec{c})$
(C) $12\vec{c} \cdot (\vec{a} \times \vec{b})$ (D) $-12\vec{b} \cdot (\vec{c} \times \vec{a})$

Q8 - 26 June - Shift 2

Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane

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of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v} \cdot \hat{j} = 7$,

then $\vec{v} \cdot (\hat{i} + \hat{k})$ is equal to :

- (A) 6 (B) 7 (C) 8 (D) 9

Q9 - 27 June - Shift 1

Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{b}| \in \{1, 2, \dots, 10\}$ is :

(A) 0

(B) 1

(C) 2

(D) 3

Space for your notes:

Q10 - 27 June - Shift 2

Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonal of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute. $|\vec{a}| = 1$ and $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$. If $\vec{c} = 2\sqrt{2}(\vec{a} \times \vec{b}) - 2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is :

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{3\pi}{4}$

Space for your notes:

Q11 - 28 June - Shift 1

If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} \cdot \vec{c} = 5$, $\vec{b} \perp \vec{c}$, then $122(c_1 + c_2 + c_3)$ is equal to _____.

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Q12 - 28 June - Shift 2

Let $\vec{a} = \alpha\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -2\hat{i} + \alpha\hat{j} + \hat{k}$, where

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$\alpha \in \mathbf{R}$. If the area of the parallelogram whose adjacent sides are represented by the vectors

$\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2 + 4)}$, then the value of

$2|\vec{a}|^2 + (\vec{a} \cdot \vec{b})|\vec{b}|^2$ is equal to

(A) 10

(B) 7

(C) 9

(D) 14

Q13 - 28 June - Shift 2

Let $\vec{a}$ be a vector which is perpendicular to the vector

Space for your notes:

$3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}$. If $\vec{a} \times (2\hat{i} + \hat{k}) = 2\hat{i} - 13\hat{j} - 4\hat{k}$, then

the projection of the vector $\vec{a}$ on the vector

$2\hat{i} + 2\hat{j} + \hat{k}$ is

(A) $\frac{1}{3}$

(B) 1

(C) $\frac{5}{3}$

(D) $\frac{7}{3}$

Q14 - 29 June - Shift 1

Let $\vec{a} = \alpha\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} - \beta\hat{j} + 4\hat{k}$ and

$\vec{c} = \hat{i} + 2\hat{j} - 2\hat{k}$ where $\alpha, \beta \in \mathbb{R}$, be three vectors. If

the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and

$\vec{b} \times \vec{c} = -6\hat{i} + 10\hat{j} + 7\hat{k}$, then the value of $\alpha + \beta$ equal to :

(A) 3

(B) 4

(C) 5

(D) 6

Space for your notes:

Q15 - 29 June - Shift 2

Let A, B, C be three points whose position vectors respectively are:

$$\vec{a} = \hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{b} = 2\hat{i} + \alpha\hat{j} + 4\hat{k}, \alpha \in \mathbb{R}$$

$$\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$$

If α is the smallest positive integer for which

$\vec{a}, \vec{b}, \vec{c}$ are non-collinear, then the length of the median, in $\triangle ABC$, through A is:

(A) $\frac{\sqrt{82}}{2}$

(B) $\frac{\sqrt{62}}{2}$

(C) $\frac{\sqrt{69}}{2}$

(D) $\frac{\sqrt{66}}{2}$

Space for your notes:

Q16 - 29 June - Shift 2

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector

Space for your notes:

such that $\vec{a} + (\vec{b} \times \vec{c}) = \vec{0}$ and $\vec{b} \cdot \vec{c} = 5$. Then, the

value of $3(\vec{c} \cdot \vec{a})$ is equal to ____.

Answer Key

Q1 (C)

Q2 (2)

Q3 (C)

Q4 (B)

Q5 (576)

Q6 (14)

Q7 (A)

Q8 (D)

Q9 (A)

Q10 (D)

Q11 (150)

Q12 (D)

Q13 (C)

Q14 (A)

Q15 (A)

Q16 (10)

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Q1 (C)

$$|\hat{b}|^2 = |\vec{c} + 2(\vec{c} \times \hat{a})|^2$$

$$|\hat{b}|^2 = |\vec{c}|^2 + 4|\vec{c} \times \hat{a}|^2 + 4\vec{c} \cdot (\vec{c} \times \hat{a})$$

$$1 = |\vec{c}|^2 + 4|\vec{c}|^2 \sin^2 \frac{\pi}{12} + 0$$

$$1 = |\vec{c}|^2 + 4|\vec{c}|^2 \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right)^2$$

$$|\vec{c}|^2 = \frac{1}{3-\sqrt{3}} = \frac{3+\sqrt{3}}{6}$$

$$\text{So } 6^2 |\vec{c}|^2 = 6(3+\sqrt{3})$$

Q2 (2)

$$\mathbf{a}_1 = (-1, 0, 3)$$

$$\mathbf{a}_2 = (0, -1, 2)$$

$$\mathbf{b}_1 = (1, -a, 0) \text{ dr's of line (1)}$$

$$\mathbf{b}_2 = (1, -1, 1) \text{ dr's of line (2)}$$

$$\bar{\mathbf{a}}_2 - \bar{\mathbf{a}}_1 = (1, -1, -1)$$

$$\bar{\mathbf{b}}_1 \times \bar{\mathbf{b}}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -a & 0 \\ 1 & -1 & 1 \end{vmatrix}$$

$$\bar{\mathbf{b}}_1 \times \bar{\mathbf{b}}_2 = \hat{i}(-a) - \hat{j} + \hat{k}(a-1)$$

$$|\bar{\mathbf{b}}_1 \times \bar{\mathbf{b}}_2| = \sqrt{a^2 + 1 + (a-1)^2}$$

$$\mathbf{a}_2 - \mathbf{a}_1 \cdot \bar{\mathbf{b}}_1 \times \bar{\mathbf{b}}_2 = 2 - 2a$$

$$\frac{2(1-a)}{\sqrt{a^2 + 1 + (a-1)^2}} = \sqrt{\frac{2}{3}}$$

Squaring on both the side

$$\text{After solving } a = 2, \frac{1}{2}$$

Q3 (C)

$$|(\hat{\mathbf{a}} + \hat{\mathbf{b}}) + 2(\hat{\mathbf{a}} \times \hat{\mathbf{b}})| = 2, \theta \in (0, \pi)$$

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$$(\vec{a} + \vec{b})^2 + 4(\vec{a} \times \vec{b})^2 + 0 = 4$$

Hints and Solutions

Let the angle be θ between $\hat{a}$ and $\hat{b}$

$$2 + 2\cos\theta + 4\sin^2\theta = 4$$

$$2 + 2\cos\theta - 4\cos^2\theta = 0$$

Let $\cos\theta = t$ then

$$2t^2 - t - 1 = 0$$

$$2t^2 - 2t + t - 1 = 0$$

$$2t(t - 1) + (t - 1) = 0$$

$$(2t + 1)(t - 1) = 0$$

$$t = -\frac{1}{2} \quad \text{or} \quad t = 1$$

$$\cos\theta = -\frac{1}{2} \quad \text{not possible as } \theta \in (0, \pi)$$

$$\theta = \frac{2\pi}{3}$$

Now,

$$S_1: 2|\vec{a} \times \vec{b}| = 2\sin\left(\frac{2\pi}{3}\right)$$

$$|\hat{a} - \hat{b}| = \sqrt{1 + 1 - 2\cos\left(\frac{2\pi}{3}\right)}$$

$$= \sqrt{2 - 2 \times \left(-\frac{1}{2}\right)}$$

$$= \sqrt{3}$$

S_1 is correct.

S_2 projection of $\hat{a}$ on $(\hat{a} + \hat{b})$.

$$\frac{\hat{a} \cdot (\hat{a} + \hat{b})}{|\hat{a} + \hat{b}|} = \frac{1 + \cos\left(\frac{2\pi}{3}\right)}{\sqrt{2 + 2\cos\frac{2\pi}{3}}}$$

$$= \frac{1 - \frac{1}{2}}{\sqrt{1}}$$

$$= \frac{1}{2}$$

C Option is true.

Q4 (B)

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\vec{a} = \lambda \left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k} \right) = \frac{\lambda}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

Now projection of $\vec{a}$ on $\vec{b} = 7$

$$\Rightarrow \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = 7$$

$$\frac{\lambda}{\sqrt{3}} \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (3\hat{i} + 4\hat{j})}{5} = 7$$

$$\lambda = 5\sqrt{3}$$

$$\vec{a} = 5(\hat{i} + \hat{j} + \hat{k})$$

$$\text{now } \vec{b} = 5\alpha(\hat{i} + \hat{j} + \hat{k}) + \beta(\hat{i})$$

$$\vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow 25\alpha(3) + 5\beta = 0$$

$$\Rightarrow 15\alpha + \beta = 0 \Rightarrow \beta = -15\alpha$$

$$\vec{b} = 5\alpha(-2\hat{i} + \hat{j} + \hat{k})$$

$$|\vec{b}| = 5\sqrt{3}$$

$$\Rightarrow \alpha = \pm \frac{1}{\sqrt{2}}$$

$$\vec{b} = \pm \frac{5}{\sqrt{2}}(-2\hat{i} + \hat{j} + \hat{k})$$

Projection of $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is

$$\frac{\vec{b} \cdot (3\hat{i} + 4\hat{j})}{5} = \pm \frac{5}{\sqrt{2}} \left(\frac{-6 + 4}{5} \right) = \pm \sqrt{2}$$

Q5 (576)

$$|\vec{a}| = 4, |\vec{b}| = 3 \quad \theta \in \left(\frac{\pi}{4}, \frac{\pi}{3} \right)$$

$$\left| (\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) \right|^2 + 4(\vec{a} \cdot \vec{b})^2$$

$$|\vec{a} \times \vec{b} - \vec{b} \times \vec{a}|^2 + 4a^2b^2 \cos^2 \theta$$

$$2|\vec{a} \times \vec{b}|^2 + 4a^2b^2 \cos^2 \theta$$

$$4a^2b^2 \sin^2 \theta + 4a^2b^2 \cos^2 \theta$$

$$4a^2b^2 = 4 \times 16 \times 9 = 576$$

Q6 (14)

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

$$\Rightarrow 13 - 1 - 4\lambda = 0 \Rightarrow \lambda = 3$$

$$\Rightarrow \vec{b} = \hat{i} + \hat{j} + 3\hat{k} \Rightarrow \vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$$

$$\Rightarrow (\vec{a} \times \vec{b}) \times \vec{b} = (13\hat{i} - \hat{j} - 4\hat{k}) \times (\hat{i} + \hat{j} + 3\hat{k})$$

$$\Rightarrow -21\hat{i} - 11\hat{j} = \hat{i} + 43\hat{j} + 14\hat{k}$$

$$\Rightarrow \vec{a} = -2\hat{i} + 2\hat{j} - 7\hat{k}$$

$$\text{Now } (\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k}) = 14$$

Q7 (A)

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = 3\vec{b} - \vec{c}$$

$$\vec{b} \times (\vec{c} \times \vec{a}) = (\vec{b} \cdot \vec{a})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a} = \vec{c} - 2\vec{a}$$

$$\vec{c} \times (\vec{b} \times \vec{a}) = (\vec{c} \cdot \vec{a})\vec{b} - (\vec{c} \cdot \vec{b})\vec{a} = 3\vec{b} - 2\vec{a}$$

$$[3\vec{b} - \vec{c}, \vec{c} - 2\vec{a}, 3\vec{b} - 2\vec{a}]$$

$$(3\vec{b} - \vec{c}) \cdot [(\vec{c} - 2\vec{a}) \times (3\vec{b} - 2\vec{a})]$$

$$(3\vec{b} - \vec{c}) \cdot [3(\vec{c} \times \vec{b}) - 2(\vec{c} \times \vec{a}) - 6(\vec{a} \times \vec{b})]$$

$$-6[\vec{b} \cdot \vec{c} \cdot \vec{a}] + 6[\vec{c} \cdot \vec{a} \cdot \vec{b}]$$

Q8 (D)

$$\vec{v} = \lambda\vec{a} + \mu\vec{b}$$

$$\vec{v} = \lambda(1, 1, 2) + \mu(2, -3, 1)$$

$$\vec{v} = (\lambda + 2\mu, \lambda - 3\mu, 2\lambda + \mu)$$

$$\vec{v} \cdot \hat{j} = 7 \quad \vec{v} \cdot \frac{\vec{c}}{|\vec{c}|} = \frac{2}{\sqrt{3}}$$

$$\lambda - 3\mu = 7 \quad \vec{v} \cdot \vec{c} = 2$$

$$\lambda + 2\mu - \lambda + 3\mu + 2\lambda + \mu = 2$$

$$2\lambda + 6\mu = 2$$

$$\lambda + 3\mu = 1$$

$$\lambda - 3\mu = 7$$

$$2\lambda = 8$$

$$\lambda = 4$$

$$\mu = -1$$

$$\text{We get } \vec{v} = (2, 7, 7)$$

Q9 (A)

Hints and Solutions

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$$\vec{a} = i + j + k$$

$$\vec{c} = 2i - 3j + 2k$$

$$\vec{b} \times \vec{c} = \vec{a}$$

$$|\vec{b}| \in \{1, 2, \dots, 10\}$$

$$\therefore \vec{b} \times \vec{c} = \vec{a}$$

$\Rightarrow \vec{a}$ is perpendicular to $\vec{b}$ as well as $\vec{a}$ is

perpendicular to $\vec{c}$

$$\text{Now } \vec{a} \cdot \vec{c} = 2 - 3 - 2 = -3 \neq 0$$

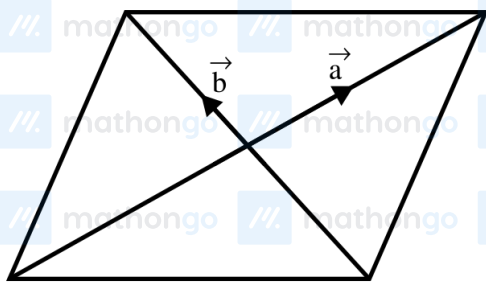
This $\vec{b} \times \vec{c} = \vec{a}$ is not possible.

$$\text{No. of vectors } \vec{b} = 0$$

Q10 (D)

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$$\text{Area} = \frac{1}{2} |\vec{a} \times \vec{b}| = 2\sqrt{2} \Rightarrow |\vec{a} \times \vec{b}| = 4\sqrt{2}$$

$$|\vec{a}| = 1 \text{ and } |\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$$

$$\Rightarrow \cos \theta = \sin \theta$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$\therefore |\vec{a} \times \vec{b}| = 4\sqrt{2} \Rightarrow |\vec{a}| |\vec{b}| \sin \frac{\pi}{4} = 4\sqrt{2}$$

$$\Rightarrow |\vec{b}| = 8$$

$$\text{Now, } \vec{c} = 2\sqrt{2} (\vec{a} \times \vec{b}) - 2\vec{b}$$

$$|\vec{c}| = \sqrt{(2\sqrt{2})^2 |\vec{a} \times \vec{b}|^2 + (2|\vec{b}|)^2} = 16\sqrt{2}$$

$$\text{Now, } \vec{b} \cdot \vec{c} = -2|\vec{b}|^2$$

$$\Rightarrow 8 \times 16\sqrt{2} \times \cos \alpha = -2.64$$

$$\Rightarrow \cos \alpha = -\frac{1}{\sqrt{2}} \Rightarrow \alpha = \frac{3\pi}{4}$$

Q11 (150)

Hints and Solutions

MathonGo

$$\bar{a} \cdot \bar{c} = 5 \Rightarrow 2c_1 + c_2 + 3c_3 = 5 \quad \dots(1)$$

$$\bar{b} \cdot \bar{c} = 0 \Rightarrow 3c_1 + 3c_2 + c_3 = 0 \quad \dots(2)$$

$$\text{And } [\bar{a} \ \bar{b} \ \bar{c}] = 0 \Rightarrow \begin{vmatrix} c_1 & c_2 & c_3 \\ 2 & 1 & 3 \\ 3 & 3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 8c_1 - 7c_2 - 3c_3 = 0 \quad \dots(3)$$

By solving (1), (2), (3) we get

$$c_1 = \frac{10}{122}, c_2 = \frac{-85}{122}, c_3 = \frac{225}{122}$$

$$\therefore 122(c_1 + c_2 + c_3) = 150$$

Q12 (D)

$$\vec{a} = \alpha \hat{i} + 2\hat{j} - \hat{k}, \vec{b} = -2\hat{i} + \alpha\hat{j} + \hat{k},$$

$$\text{area of parallelogram} = |\hat{a} \times \hat{b}|$$

$$|\hat{a} \times \hat{b}| = \sqrt{(\alpha + 2)^2 + (\alpha - 2)^2 + (\alpha^2 + 4)^2}$$

$$\text{Given } |\hat{a} \times \hat{b}| = \sqrt{15(\alpha^2 + 4)}$$

$$2(\alpha^2 + 4) + (\alpha^2 + 4)^2 = 15(\alpha^2 + 4)$$

$$(\alpha^2 + 4)^2 = 13(\alpha^2 + 4)$$

$$\Rightarrow \alpha^2 + 4 = 13 \therefore \alpha^2 = 9$$

$$2|\vec{a}|^2 + (\vec{a} \cdot \vec{b})|\vec{b}|^2$$

$$|\vec{a}|^2 = \alpha^2 + 4 + 1 = \alpha^2 + 5$$

$$|\vec{b}|^2 = 4 + \alpha^2 + 1 = \alpha^2 + 5$$

$$\vec{a} \cdot \vec{b} = -2\alpha + 2\alpha - 1 = -1$$

$$\therefore 2|\vec{a}|^2 + (\vec{a} \cdot \vec{b})|\vec{b}|^2$$

$$2(\alpha^2 + 5) - 1(\alpha^2 + 5) = \alpha^2 + 5 = 14$$

Q13 (C)

$$\begin{aligned}
 & (\vec{a} \times (2\hat{i} + \hat{k})) \times \left(3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}\right) \\
 &= (2\hat{i} - 13\hat{j} - 4\hat{k}) \times \left(3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}\right) \\
 &= -(6+2)\vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -13 & -4 \\ 3 & \frac{1}{2} & 2 \end{vmatrix} \\
 &\vec{a} = 3\hat{i} + 2\hat{j} - 5\hat{k}
 \end{aligned}$$

Projection of $\vec{a}$ on vector $2\hat{i} + 2\hat{j} + \hat{k}$ is

$$\vec{a} \cdot \frac{(2\hat{i} + 2\hat{j} + \hat{k})}{3} = \frac{5}{3}$$

Q14 (A)

$$\begin{aligned}
 \frac{\vec{a} \cdot \vec{c}}{|\vec{c}|} &= \frac{10}{3} \\
 \Rightarrow \frac{\alpha + 6 + 2}{\sqrt{1+4+4}} &= \frac{10}{3} \Rightarrow \alpha = 2
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -\beta & 4 \\ 1 & 2 & -2 \end{vmatrix} &= -6\hat{i} + \hat{j} + \hat{k} \\
 \Rightarrow 2\beta - 8 &= -6 \Rightarrow \beta = 1 \\
 \Rightarrow \alpha + \beta &= 3
 \end{aligned}$$

Q15 (A)

$$\vec{AB} \parallel \vec{AC} \text{ if } \frac{1}{2} = \frac{\alpha - 4}{-6} = \frac{1}{2} \Rightarrow \alpha = 1$$

$\vec{a}, \vec{b}, \vec{c}$ are non-collinear for $\alpha = 2$ (smallest positive integer)

$$\text{Mid-point of BC} = M\left(\frac{5}{2}, 0, \frac{9}{2}\right)$$

$$AM = \sqrt{\frac{9}{4} + 16 + \frac{9}{4}} = \frac{\sqrt{82}}{2}$$

Q16 (10)

$$\vec{a} + \vec{b} \times \vec{c} = 0$$

$$\vec{a} \times \vec{b} + |\vec{b}|^2 \vec{c} - 5\vec{b} = 0$$

$$\text{It gives } \vec{c} = \frac{1}{3}(10\hat{i} + 3\hat{j} + 2\hat{k})$$

$$\text{so } 3\vec{a} \cdot \vec{c} = 10$$

But it does not satisfy $\vec{a} + \vec{b} \times \vec{c} = 0$.

This question has data error.

Alternate (Explanation) :

According to given $\vec{a}$ & $\vec{b}$

$$\vec{a} \cdot \vec{b} = 1 - 2 + 3 = 2 \quad \dots (i)$$

but given equation

$$\vec{a} = -(\vec{b} \times \vec{c})$$

$$\Rightarrow \vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$$

which contradicts.